

Caribbean Coral Reef Institute (CCRI) Water Quality Monitoring Lab University of Puerto Rico – Mayagüez Campus Department of Marine Sciences			
Document Title	Document Number	Version	
SOP Enterococcus	WQP 009	1.1	

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VERSION HISTORY			
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1.0	5/10/23	Initial Release	TAC
1.1	9/11/23	Analyze samples within 6 hours instead of 8 hours.	TAC

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1. Title

Standard Operating Procedure for *Enterococcus* Quantification.

2. Applicability

This SOP applies to the determination of *Enterococcus* content at shallow coral reef sites around Puerto Rico.

3. Purpose

- 3.1. Provide laboratory-specific instructions for the CCRI Water Quality Monitoring Lab personnel to analyze coral reef seawater *Enterococcus* content at shallow coral reef sites around Puerto Rico.
- 3.2. Provide water quality data to be uploaded to the EPA WQX database for use by state and federal agencies and any other parties interested in coral reef seawater *Enterococcus* content data following the Quality Assurance Project Plan.

4. Method

Standard Method 9230-D (Baird et al., 2017).

5. Health and Safety

- 5.1. Wear clean lab gloves and use sterile pipettes throughout while performing this SOP.
- 5.2. Always wear a lab coat and safety glasses while preparing reagents and performing this SOP.
- 5.3. Use an UV light lamp with the appropriate precautions and eye protection.
- 5.4. The sealer device is a burn hazard if not properly used and maintained. Therefore only trained personnel should manage and/or clean the sealer.
- 5.5. Refer to the Safety Data Sheets (SDS) for all chemicals used in this SOP.

6. General Precautions

- 6.1. Always maintain two points of contact when deploying the Van Dorn bottle on the boat to avoid losing the bottle overboard.
- 6.2. Ensure the boat engines are idle and the captain approves of the Van Dorn bottle being deployed overboard before deploying the bottle to ensure that the rope does not become entangled with the boat propeller.
- 6.3. Do not use solvents, alkaline or abrasive cleaners or other abrasives such as steel wool or scouring powder.
- 6.4. Oil and grease may interfere with the results.
- 6.5. High mineral content may interfere with the results.
- 6.6. Do not remove the outer shell of the sealer or spill water into the sealer.
- 6.7. Personal injury, sealer damage, property damage, and/or inaccurate test results may occur if sealer is not used according to the manufacturer instructions.
- 6.8. Applying labels of any kind to trays before sealing may damage the sealer. Use a felt-tip permanent marker to label the trays.
- 6.9. Do not reverse the motor of the sealer if the rubber insert has been drawn fully into the input slot.
- 6.10. Do not open the access doors of the sealer or tilt the sealer if sample is dripping from the unit.

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Wait until dripping stops and the sealer is cool.

6.11. Do not touch the upper roller of the sealer if it is hot.

7. Training

- 7.1. Personnel shall be trained in the proper sample collection, handling, and analysis procedure for *Enterococcus* quantification.
- 7.2. Personnel shall receive training on how to complete the field sampling log for this SOP.

8. Required Materials

- 8.1. Field Sampling Log (Appendix F)
- 8.2. Enterococcus lab notebook
- 8.3. IDEXX Enterolert Test
- 8.4. IDEXX Sterile Quanti-Tray/2000
- 8.5. IDEXX 120 mL Sterile Sample Vessel
- 8.6. Quanti-Tray Sealer Plus with rubber inserts
- 8.7. Fisherbrand Mini Low Temperature Refrigerated Incubator
- 8.8. UV lamp (365 to 366 nm)
- 8.9. Biohazard waste bags
- 8.10. Whirl-Pak® bags
- 8.11. HOBO temperature sensor to monitor incubator temperature
- 8.12. KWIK-STIK Bacteria Packages (3 bacteria strains)
- 8.13. Millipore Type 1 ultrapure water
- 8.14. Inoculating loop
- 8.15. Bunsen burner
- 8.16. Incubator
- 8.17. Tryptic Soy Agar (TSA) plates (3)
- 8.18. Ethanol 70% for disinfecting

9. Procedure

9.1. Sample Collection

- 9.1.1. Prepare a Van Dorn bottle to be deployed.
 - 9.1.1.1. A fresh Van Dorn bottle must be used for each sampling location.
 - 9.1.1.2. Each Van Dorn bottle will be rinsed with chlorinated tap water following each sampling day and allowed to air dry before collecting the next sample to minimize any potential Enterococcus cross contamination between sample sites.
- 9.1.2. Lower the Van Dorn bottle in a slow and steady mode until it gets close to the bottom with caution not to touch the bottom and stir sediment up.
- 9.1.3. Drop the bottle messenger to close the Van Dorn bottle at the desired depth.
- 9.1.4. Pull the Van Dorn bottle back up into the boat.
- 9.1.5. Push open the Van Dorn bottle's spigot valve to allow water flow.
- 9.1.6. Collect the sample in a Whirl-Pak® bag using a new pair of gloves.
 - 9.1.6.1. Avoid touching the inside of the bag with the spigot valve or anything else.
 - 9.1.6.2. Refer to Appendix J for the seawater sample collection order.
- 9.1.7. Close the bag by twisting the upper side and fixing it with the wire band.

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9.1.8. Place the sample bag inside the cooler until analysis in the lab.

9.1.9. Process samples within 6 hours.

9.2. Sample Preparation & Analysis

9.2.1. All procedures should be conducted wearing sterile gloves to minimize cross-contamination of bacteria into samples.

9.2.2. Turn ON the IDEXX Quanti-Tray sealer to pre-heat by pressing the switch on the back of the sealer.

9.2.2.1. Wait for the sealer to heat and display a green status light.

9.2.3. Ensure the incubator is at $41 \pm 0.5^{\circ}\text{C}$.

9.2.4. Prepare IDEXX sterile vessels for sample addition.

9.2.4.1. Set apart the IDEXX sterile vessels to be used according to the total of samples to be analyzed.

9.2.4.2. Label vessels with sample identification number.

9.2.4.3. Add 90 mL of Millipore water to each IDEXX sterile vessel.

9.2.4.4. Carefully crack open the Enterolert media packages and empty one into each vessel.

9.2.4.5. Close the vessels and invert them gently until the reagent is fully dissolved.

9.2.4.6. Note: Sample must be added to the vessels within 30 mins.

9.2.5. Pipette 10 mL of the sample from the Whirl-Pak® bag into the corresponding IDEXX vessel.

9.2.6. Close the IDEXX vessel and gently agitate to ensure sample mixes with the media.

9.2.7. Carefully open a Quanti-Tray as per the manufacturer's instructions.

9.2.7.1. Hold the Quanti-Tray upright with the well side facing the palm of your hand.

9.2.7.2. Squeeze the upper part of the Quanti-Tray so that it bends toward the palm of your hand.

9.2.7.3. Pull the tab gently outwards and grip around the plastic side of the tray with one hand, pressing slightly to open it wider.

9.2.8. Pour the entire contents of the IDEXX vessel into the Quanti-Tray.

9.2.8.1. Avoid contact with the foil tab.

9.2.8.2. Pour slowly to avoid creating bubbles.

9.2.9. Gently tap out big bubbles before sealing.

9.2.10. Insert the Quanti-Tray into the sealer following manufacturer instructions.

9.2.10.1. Place the tray in the sealer well-side down on to the rubber insert.

9.2.10.2. Guide the loaded rubber insert into the input slot of the sealer, with the large cutout facing away from the sealer.

9.2.10.3. Slide the rubber insert into the sealer until the unit begins to pull it. .

9.2.10.4. Press and hold the "Reverse" button (↩) to stop a misaligned tray.

9.2.11. Remove the rubber insert and tray.

9.2.11.1. The insert and tray will come out from the sealer's back.

9.2.11.2. Turn off the sealer or use the "sleep mode" ("Moon" button) when not in use.

9.2.12. Fill out sample information on the back of the tray.

9.2.12.1. With a permanent marker (so as not to tear the paper on the back of the Quanti-Tray) gently mark the Quanti-Trays with the lab identification number of each sample.

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9.2.13. Incubate at $41 \pm 0.5^{\circ}\text{C}$ for 24 hours.

9.2.13.1. Record the time and temperature before incubation for the respective sample in the enterococcus lab notebook

9.3. Evaluating Most Probable Number (MPN)

9.3.1. After the incubation period (24 hours), remove the tray from the incubator.

9.3.1.1. Record the time and temperature after incubation for the respective sample in the enterococcus lab notebook.

9.3.1.2. The incubation time must be at least 24 hours and not more than 28 hours.

9.3.2. Place the tray under the UV light and count the number of large and small wells with a bright, blue fluorescence.

9.3.2.1. Avoid direct eye contact with the lamp.

9.3.2.2. Do not place the lamp too close to the sample.

9.3.3. Mark fluorescing wells with a permanent marker to keep track of accounted wells.

9.3.4. Record the number of positive large wells and positive small wells in the enterococcus lab notebook.

9.3.5. Use the MPN (most probable number) chart provided with the Enterolert test kit to determine the amount of enterococcus per 100 mL sample:

9.3.5.1. Lack of fluorescence = Negative for *enterococci*

9.3.5.2. Blue fluorescence = Positive for *enterococci*

9.3.6. Record the MPN for the respective sample in the enterococcus lab notebook.

9.3.7. The final sample MPN should be corrected for the 1:10 dilution by multiplying the resulting MPN by 10. Record this dilution corrected MPN as a separate value.

9.3.8. Discard Quanti-Tray wells as biohazard.

9.3.8.1. Decontaminate prior to disposal by autoclaving at 121°C for 60 minutes

10. Documentation

10.1. Field data will be recorded in the Sampling Log (Appendix F).

10.2. Field form shall be kept in the designated file cabinet at CCRI office.

10.3. Results for Enterococcus analysis will be recorded in the Enterococcus lab notebook.

10.4. Data contained in hard copy forms shall be digitized into the corresponding electronic spreadsheets as soon as possible after the task is completed.

10.5. All data will be electronically archived and backed up online.

11. Cleaning and Maintenance

11.1. Check the Quanti-Tray sealer and inserts before each use and refer to Quanti-Tray User Manual for part-specific cleaning instructions.

11.1.1. Outside of the sealer

11.1.2. Inside of the sealer

11.1.3. Rubber inserts

12. Troubleshooting

12.1. Refer to pages 4-5 of the Quanti-Tray Sealer user manual for most common troubleshooting instructions.

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- 12.2. For complex troubleshooting or repairs other than cleaning, contact IDEXX Technical Service at +1 800 321 0207.

13. QA/QC

- 13.1. As per QAPP Table 4, seawater *Enterococcus* concentration should be measured as ind/100mL with a field duplicate precision of <60%.
- 13.2. Any seemingly erroneous measurements will be flagged for further review in accordance with the QAPP.
- 13.3. One external reference sample must be analyzed per year.
- 13.4. Verification of Enterolert Media
- 13.4.1. Bacterial Inoculation Procedure
- This inoculation will be made for the following bacterial strains: *Serratia marcescens*, *Aerococcus viridans*, and *Enterococcus faecium*.
- 13.4.1.1. Perform this process inside the Biosafety cabinet level 2, and use lab coat and sterile gloves.
- 13.4.1.2. Clean the surface area with ethanol 70 %.
- 13.4.1.3. Remove from the fridge the three Tryptic Soy Agar (TSA) media plates and the three bacteria strains KWIK-STIK Kits, and place them inside the Biosafety cabinet.
- 13.4.1.3.1. Let the media and the KWIK-STIK kits reach ambient temperature.
- 13.4.1.4. Open a TSA media plate for the bacteria strain to inoculate.
- 13.4.1.5. Tear open one KWIK-STIK pouch and remove the unit.
- 13.4.1.6. Tear off the pull-tab portion on the label and attach it to the bottom of the TSA plate .
- 13.4.1.7. Crack the ampule at the top of the unit (just below the fluid meniscus).
- 13.4.1.8. Hold the unit vertically and tap on a hard surface.
- 13.4.1.8.1. Do not shake the unit.
- 13.4.1.8.2. Do not cover the vent hole on top of the unit.
- 13.4.1.9. Gently pinch the bottom of the unit several times to help dissolve the pellet.
- 13.4.1.10. Invert the unit and saturate the swab with the hydrated material.
- 13.4.1.11. Inoculate the culture by rolling the swab over $\frac{1}{3}$ of the petri dish.
- 13.4.1.12. Sterilize the inoculating loop by holding it in a Bunsen burner flame until it is red hot.
- 13.4.1.13. Touch a small empty surface of the media in the TSA plate with the inoculating loop until it stops sizzling.
- 13.4.1.14. Continue streaking the plate using the sterile inoculation loop.
- 13.4.1.15. Discard the KWIK-STIK unit in a biohazard bag.
- 13.4.1.16. Cover the inoculated plate and set apart.
- 13.4.1.17. Repeat steps 13.3.1.4 to 13.3.1.16 for the remaining two bacteria strains.
- 13.4.1.18. Incubate the inoculated plate at $35^{\circ}\text{C} \pm 0.5^{\circ}\text{C}$ for 18 to 24 hours.
- 13.4.1.18.1. Record temperature before and after incubation in the Enterococcus lab notebook and verify temperatures were maintained throughout incubation using the HOBO temperature sensor located inside the incubator.

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13.4.1.19. After the incubation period is over, store the TSA media plates containing the bacteria colonies upside down, inside of a plastic bag, at 4°C.

13.4.2. Verification Analysis

This verification will be performed for each batch of Enterolert test kit.

13.4.2.1. Prepare four IDEXX vessels.

13.4.2.1.1. Label vessels: (“B” for Blank, “P” for Positive, and two vessels “N” for Negative)

13.4.2.1.2. Pour 100mL of Millipore water into each vessel.

13.4.2.1.3. Cap vessels and set apart for future use.

13.4.2.2. Prepare test tubes for water inoculation.

13.4.2.2.1. Label one test tube for each of the bacteria strains (*Serratia marcescens*, *Aerococcus viridans*, *Enterococcus faecium*)

13.4.2.2.2. Pour 5mL of Millipore water into each glass test tube.

13.4.2.3. Prepare Blank Control

13.4.2.3.1. Use IDEXX vessel “B” prepared on step 13.3.3.1.

13.4.2.4. Prepare Negative Controls

13.4.2.4.1. Sterilize an inoculating loop using the Bunsen burner

13.4.2.4.2. Touch a colony of *Serratia marcescens* from the plate with the sterile loop.

13.4.2.4.3. Insert the loop containing the bacteria into the corresponding previously labeled test tube and gently shake the loop in the water.

13.4.2.4.4. Remove the loop, cap and gently shake the test tube to thoroughly mix.

13.4.2.4.5. Reopen the test tube and touch the water with the previously used inoculating loop.

13.4.2.4.6. Transfer the loop to the IDEXX vessel with label “N” and gently shake the loop.

13.4.2.4.7. Remove the loop, close the vessel and gently invert to mix contents.

13.4.2.4.8. Repeat steps 13.3.4.4.1 to 13.3.4.4.7 for *Aerococcus viridans*.

13.4.2.5. Prepare Positive Control

13.4.2.5.1. Sterilize an inoculating loop using the Bunsen burner.

13.4.2.5.2. Touch a colony of *Enterococcus faecium* from the plate with the sterile loop.

13.4.2.5.3. Insert the loop containing the bacteria into the corresponding previously labeled test tube and gently shake the loop in the water.

13.4.2.5.4. Remove the loop, cap and gently shake the test tube to thoroughly mix.

13.4.2.5.5. Reopen the test tube and touch the water with the previously used inoculating loop.

13.4.2.5.6. Transfer the loop to the IDEXX vessel with label “P” and gently shake the loop.

13.4.2.5.7. Remove the loop, close the vessel and gently invert to mix contents.

13.4.2.6. Prepare and analyze four Quanti-Trays, one for each control, as described

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starting at step 9.2.7.

13.4.2.7. Evaluate MPN as for regular samples following steps on Section 9.3.

14. References

- 14.1. Baird, R. B., Eaton, A. D., & Rice, E. W. (2017). Standard methods for the examination of water and wastewater (Vol. 23). E. W. Rice (Ed.). Washington, DC: American Public Health Association.
- 14.2. IDEXX Enterolert Manual.
- 14.3. Quanti-Tray sealer Manual.